

OPTIMIZATION OF TRAFFIC LIGHTS AT CROSSROADS

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We study the influence of traffic lights on the traffic flow in cities. The urban traffic is simulated in the cellular automata framework. Both the deterministic and probabilistic models are discussed. The effects of speed limit and stochastic noise are analyzed. The operation of a traffic light is characterized by two parameters: signal period and phase allocation. With two traffic lights on road, one more parameter is prescribed: synchronization shift. The results of tuning these parameters are presented in the fundamental diagram. We examine the traffic flow and discuss the choice of optimized setting in different density regions.

Keywords: Cellular automata; Nagel–Schreckenberg model; traffic flow; traffic signal control.

1. Introduction

Traffic signal control has always been an important part of urban traffic management. Due to the crisscross structures of urban streets, vehicles traveling in crossing streets have to take turns passing through street intersections. The timing and duration of traffic lights play the role of assigning passing priority that allows or disrupts vehicular flow, and has been one of the most effective methods of regulating urban traffic flow. The main goal of the present paper is to adjust the traffic lights such that vehicles crossing in different directions meet minimal red light delay and are able to maintain a high traffic flow and low travel time. For a fixed street structure, given a specified traffic cycle period, phase allocation and vehicle speed, it is desirable to synchronize signals to achieve high traffic flow with maximum bandwidth. The result has also been applied to real city traffic control.^{1,2} Traditionally the technique for investigating such problem has been by deterministic optimization process or recently by fuzzy logic control.^{3–5} But these methods overlook the subtle dependence of real traffic situations such as driver behavior and vehicle interaction, and the subsequent phase transition as the vehicle density varies has never been considered. To take into account the real traffic subtleties, cellular automata (CA) has been one of the most intuitive and efficient methods.⁶ The Binham–Middleton–Levine (BML) model for city traffic⁷ and Nagel–Schreckenberg (NaSch) model for

highway traffic⁸ can be combined for a realistic description of urban traffic flow.^{9–12} We will first consider the case of one single traffic light and the consequent effect on traffic flow. Two traffic lights with adjustable synchronization shift will also be examined.

2. A Single Traffic Light

First, we study the influence of an isolated traffic light. To simplify the configuration, the system consists only of a single traffic light setting up on a single-lane road. With the enforced periodic boundary condition, the exact location of the traffic light is irrelevant. The general behavior of vehicles is prescribed by the well known cellular automata traffic model, the NaSch model. There are only two parameters in the model: the speed limit $v_{\max}$ and the stochastic braking probability p . In the special case of $v_{\max} = 1$ and $p = 0$, the model reduces to the asymmetric simple exclusion processes (ASEP), i.e., vehicle hops forward to the next site as long as the site is empty. To simulate the urban traffic, the lattice site L reflects the distance between two traffic lights in the down town area. Thus a small lattice is employed. Typically a lattice of $L = 100$ corresponds to a distance of 750 m. A selected link between two neighboring sites is designated as the traffic light, which switches periodically to red and green. In the green phase, vehicles can pass the link without hindrance; in the red phase, no vehicle is allowed to pass the link, i.e., vehicles have to stop in front of the link and wait until the traffic light switches to the next green phase. Among the vehicles, overtaking is forbidden. The operation of the traffic light is characterized by two parameters: the signal period T and the ratio allocated to green phase α . The period T is understood as the duration of a green phase and a red phase. The ratio $\alpha = 0.5$ implies that the signal period is divided equally between the two phases, which is applied in the equitable intersection. When the intersection involves a main road and a sideways, the traffic light on the main road is expected to have a ratio $\alpha > 0.5$.

The typical results are shown in Fig. 1, where the parameters of the vehicular behavior are set to ASEP and a ratio $\alpha = 0.5$ is assumed. In contrast to the conventional two-dimensional plot of fundamental diagram, a three-dimensional plot of the traffic flow J as a function of both the vehicular density ρ and the signal period T is employed. Three distinct regions are observed: low-density oscillations, intermediate-density plateau, and high-density oscillations. In the low density region, the flow increases monotonically with the increase of density. As the signal period varies, an interesting oscillation of traffic flow is noticed. When the period is short, $T < 100$ s, the prominent feature is oscillation. On the other end, the oscillations subside as the signal period prolongs. Similar oscillations are also observed in the high density region. In fact, the exact symmetry between the low density and high density regions is expected in ASEP. Analytically such oscillations can be understood as the time required for the vehicles returning to meet the traffic light in green phase, $T = L/n$, n being integer, because the low density free flow and

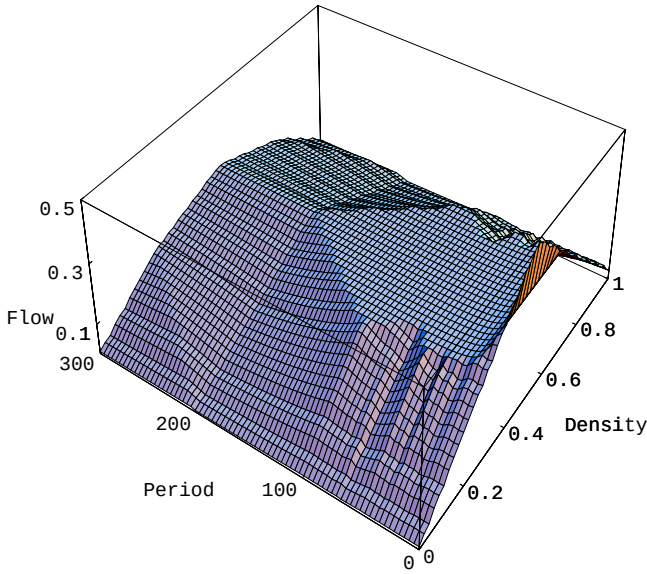


Fig. 1. Traffic flow as functions of density and signal period. The parameters are $v_{\max} = 1$, $p = 0$, and $\alpha = 0.5$.

high density jamming backflow have the same speed 1. In the intermediate density region, a plateau is present where the traffic flow is saturated to its capacity. When the signal period is short, $T < 100$ s, the plateau has wiggling boundaries which can be related to the prominent oscillations in both the low density region and the high density region. When the signal period is prolonged, $T > 100$ s, the plateau shrinks rapidly which is then related to the subsiding oscillations in both the low density region and the high density region. When the period is increased to over 200 s, the full capacity of traffic flow can only be maintained at a unique density $\rho = 0.5$ and the plateau shrinks to a straight line. We note that the reflect symmetry along the line $\rho = 0.5$ is an artifact of $v_{\max} = 1$. We also notice that the traffic flow increases with the decrease of signal period when T is very small. In practice, the traffic flow is fully restored in the limit $T \rightarrow 0$. Though it is not realistic, the behavior in this limit is easy to understand. When the signal period becomes very short, the perturbation to the traffic flow is limited. It is like an idealized stop sign and the traffic flow remains the same as in the case without the traffic light.

The oscillation pattern reflects basically the deterministic feature of the vehicular behavior. With the set in stochastic noise, the oscillations are diminished. The traffic flow are mainly determined by the vehicular density and becomes insensitive to the signal period especially when the period is large. It is interesting to note that the introduction of stochastic noise leads to significant reduction of the traffic flow, especially in the intermediate density region. Such effect enhances with increase of signal period. Thus the plateau presents a slow decreasing height with increasing period. The typical results for $p = 0.5$ are shown in Fig. 2. In the case of $v_{\max} = 1$,

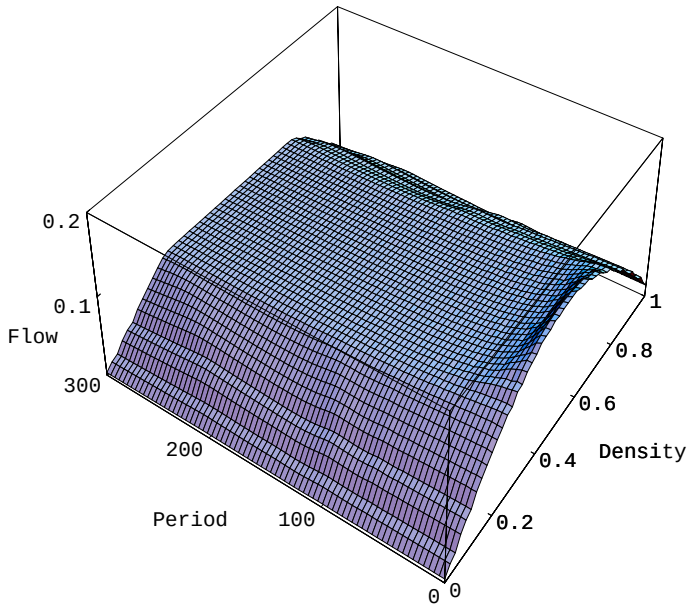


Fig. 2. Traffic flow as functions of density and signal period. The parameters are $v_{\max} = 1$, $p = 0.5$, and $\alpha = 0.5$.

the maximum flow is expected at the density $\rho = 0.5$. The operation of a traffic light leads to the saturation of traffic flow in the intermediate density region. Without the stochastic noise, such saturation is observed with a short signal period. With the noise, the flow saturates to a lower value which is then observed more clearly with a long signal period.

Increasing the speed limit would surely increase the traffic flow. The capacity, i.e., the height of the plateau, is expected to rise and shift toward the low density region. The effects of a higher speed limit, $v_{\max} > 1$, can also be observed as the asymmetry between the low density region and the high density region, because the low density free flow average speed $\langle v \rangle = v_{\max} - p$ and high density jamming backflow speed $1 - p$ are no longer equal. With the increase of $v_{\max}$, the low density region shrinks with the expansion of the high density region. The typical results are shown in Fig. 3. The oscillations in both regions can still be observed. In the low density region, much smaller amplitudes of oscillations are observed, which are then confined to the rapidly shrunk plateau observed around $T \sim 50$ s. In contrast, the rapidly shrunk plateau in the high density region is observed around $T \sim 150$ s. When the signal period is large enough, the plateau shrinks to a straight strip. The boundary between the plateau and the oscillations appears to be a straight line along a definite vehicular density. To the low density side, the straight line appears when $T > 60$; to the high density side, the straight line does not appear until $T > 240$ s. When the signal period is short, the traffic flow in the middle density region increases monotonically with the decrease of signal period in the

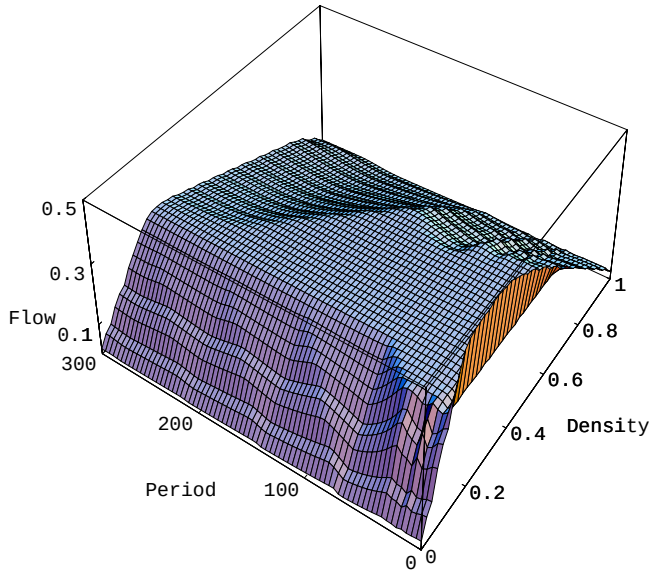


Fig. 3. Traffic flow as functions of density and signal period. The parameters are $v_{\max} = 3$, $p = 0.1$, and $\alpha = 0.5$.

case of $v_{\max} = 1$. In the case of $v_{\max} > 1$, when T decreases, the traffic flow decreases slightly before increases to nearly restore the full capacity without the traffic light.

The traffic condition is often summarized by the fundamental diagram, i.e., the relationship between the traffic flow J and the vehicular density ρ . Up to now, we explore only the influence of three parameters: the speed limit $v_{\max}$, the stochastic noise p , and the signal period T . The first two parameters are prescribed in the NaSch cellular automata to model the vehicular behavior; the parameter T is the only control parameter in the operation of a single traffic light. In the following, we will study the influence of the other two parameters: the ratio of green phase α and the lattice size L . Basically these two parameters reflect the infrastructure of urban road network, e.g., the configuration of an intersection between main road and sideways, and the distance between two intersections. The flow enhances with the increase of ratio α . As more time is allocated to the green phase, the increase of the traffic flow is expected. However, we are more concerned with the domain evolution of the three distinct regions. The domain of the low-density oscillations is unaffected by varying the ratio α , and the boundary between the plateau and the high-density oscillations moves almost linearly with the parameter α . When the ratio α decreases, the boundary moves toward the high-density end, and the plateau evolves by expanding its domain but decreasing the height. When the ratio α increases, the boundary shifts toward low density, and the plateau is then presented with a decreasing domain but an increasing height. It is interesting to observe that the boundary shifting is along the density axis. The pattern of oscillations is still

present in the same scale of the signal period. Thus the effects of the green ratio α can be simply observed as an approximate scaling relation in the variable $(\alpha\rho)$; a rescaling in the flow axis is also expected.

Similarly, a scaling interpretation can also be found to the parameter L . The very same feature of traffic flow can be observed on a larger lattice with a longer signal period. As the average vehicular speed is independent of the lattice size, a scaling relation is expected in the variable (T/L) . In summary, three distinct domains are observed on the 3D plot: J versus ρ and T . With the scaled axes $(\rho\alpha)$ and (T/L) , the domain boundaries are characterized only by $v_{\max}$ and p , i.e., the two parameters modeling the vehicular behavior.

3. Control Strategy

The operation of a traffic light is aimed to regulate the traffic and maintain the highest flow on road. In the case of a single traffic light, the only control parameter is the signal period T . The three-dimensional extension of the fundamental diagram $J(\rho, T)$ summarizes all the needed information. As the results are mainly determined by the vehicular behavior, the setting of the model parameters is important. When the NaSch model is applied to the highway traffic, the vehicular behavior can be well described by the setting: $v_{\max} = 5$ and $p = 0.5$. In contrast, the urban traffic should be characterized by a much lower speed limit. Also the stochastic noise is expected to be suppressed significantly as the vehicles driving in the city present much greater uniformity. A reasonable setting for the urban traffic would be $v_{\max} = 3$ and $p = 0.1$, see Fig. 3.

The height of the plateau denotes the capacity of the road. The precise value of the maximum flow is determined by the driving behavior and the configuration of the crossroad, which are parameterized by $v_{\max}$, p , and α . When the vehicular density is around the intermediate density region, the highest flow can be easily maintained. The traffic flow is insensitive to the variation of the signal period, as long as it is within the plateau region. Compared to the traffic flow without the hindrance of a traffic light, the capacity is reduced by a factor of α . Such reduction of traffic flow is acceptable, as the crossroad cannot open to all directions at the same time. However, without the traffic light, the maximum flow can only be achieved at a unique vehicular density. With the regulation of a traffic light, the maximum flow can be sustained over quite a range of vehicular density. If we consider the achieved capacity among different directions, the full road capacity can be achieved at a range of density. Thus, the operation of a traffic light plays the role of keeping a high efficiency of road sharing among vehicles, and is able to maintain the full road capacity over a wider range of densities.

Outside the plateau region, either in the low density region or in the high density region, the observed oscillations imply that the traffic flow is very sensitive to the signal period. To keep the flow at its highest possible value, a careful setting of the signal period is required. The oscillations subside slowly as the signal period

increases. The good thing for the control strategy is that there are many optimal values of signal period to maintain the relatively high flow. The bad thing is that the range for these optimal values is quite narrow, which means a slight deviation will lead to a notable decreasing in the traffic flow. The oscillations also come with a clear cutoff, which can also be observed as the rapid shrinkage of the plateau with the increase of signal period. Just before the cutoff, the last peak of oscillation is well separated from the other peaks, which means the last peak can provide a much wider period range for the high traffic flow. If the location of the last peak was selected as the optimal signal period, the traffic flow will be less sensitive to the deviation of setting. In both the low and high density regions, the setting can be chosen as the time required for the vehicle fronts to travel one full close loop, i.e., $L/\langle v \rangle$. From Fig. 3, a setting around $T \sim 35$ s is appropriate in the low density region; in the high density region, the appropriate setting becomes $T \sim 120$ s. When the traffic light operates in these optimal signal periods, the traffic flow can be restored to or even surpass the capacity without traffic light at the corresponding density. We further point out an important result that it is unnecessary to fine-tune the optimal signal period with the variation of density. The peaks of the oscillations are observed to locate along the density axis. Thus the optimal signal periods are optimal not just for a particular density but for the entire low density or high density region.

In summary, the traffic light should operate at only two different signal periods according to the vehicular density: one for the low density region, the other for the high density region. These two signal periods should be carefully calibrated to the driving behavior (modeled by $v_{\max}$ and p) and the distance between traffic lights (modeled by L). The signal period in the high density setting is usually the larger one. As to the switching between these two modes, it is not critical. Within the plateau region of intermediate densities, the saturated traffic flow can be achieved by both settings.

4. Two Traffic Lights

Next, we add one more traffic light on road. On a ring of L sites, two traffic lights were set up with a separation of $L/2$. Such configuration can be interpreted either as two posts separated as far as possible or as two uniformly distributed posts. With another traffic light on road, one more control parameter is prescribed: the synchronized shift or delay δ , which is understood as the time delay between these two traffic lights regarding the starting of green phase. As the two parameters $v_{\max}$ and p are modeling the vehicular behavior, and the two parameters L and α reflecting the road network, we will explore the influence of the two control parameters T and δ . The two traffic lights are assumed to operate with the same period T and the same ratio α . With such system configuration, a naive symmetry between δ and $-\delta$ is expected.

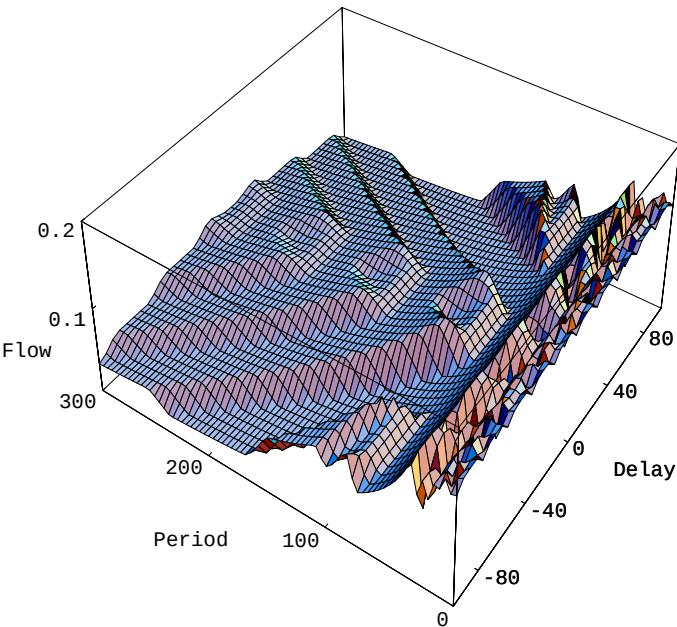


Fig. 4. Traffic flow as functions of signal period and delay at low density $\rho = 0.05$. The parameters are $v_{\max} = 3$, $p = 0.1$, and $\alpha = 0.5$.

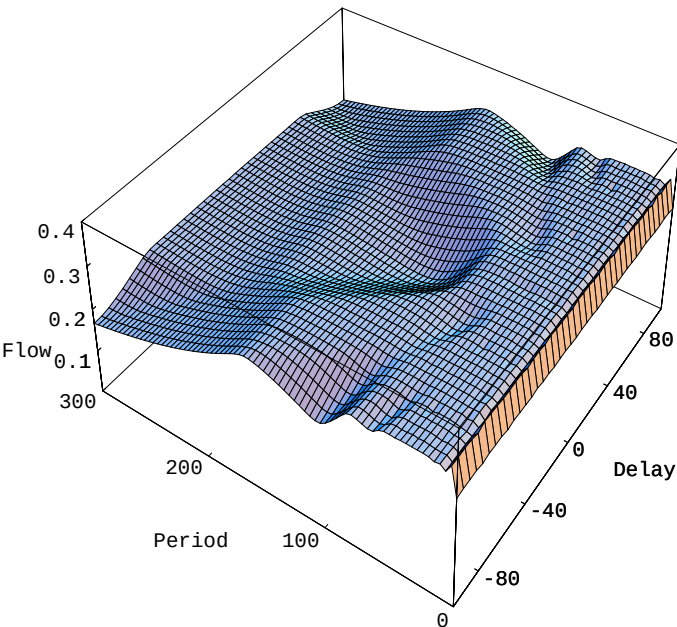


Fig. 5. Traffic flow as functions of signal period and delay at high density $\rho = 0.5$. The parameters are $v_{\max} = 3$, $p = 0.1$, and $\alpha = 0.5$.

As expected, the traffic flow is sensitive to the variation of both the period T and the delay δ . The typical results for the low vehicular density and for the intermediate density are shown in Figs. 4 and 5, respectively. When the vehicular density is low, wave-like structure can be observed as the parameters T and δ vary. To maintain the traffic flow at the peak of oscillations, a linear relationship between the period T and the delay δ is required. With a short delay, the traffic flow can be maintained at the same value as in the case of a single traffic light, i.e., the second traffic light will not further obstruct the traffic. Such appropriate delay is independent of the adapted signal period. Of course, a shorter signal period will sustain a higher traffic flow in the low density region. The highest flow can be achieved with a short signal period and a short delay. As the oscillations are observed, both setting are critical. Only a limited range for both parameters are provided.

As the vehicular density increases, the orderly wave-like structure disappears. Instead, a larger crater-like structure emerges at the center of Fig. 5, which corresponds to the setting of a long signal period and no delay. Compared to the result of one signal traffic light, the setting of the signal period is still within the plateau region, just before the cutoff to the high density oscillations. In such case, the setting of the delay becomes very critical. When the delay is set to zero, i.e., the two traffic lights switched simultaneously, the traffic condition is the worst. However, with an appropriate delay, the traffic flow can be brought to its maximum with the very same setting of the signal period. Again, the traffic flow can then be restored to the capacity of a signal traffic light. It is interesting to note that in contrast to the low density region, the setting of the delay becomes irrelevant as the signal period becomes very large. However, as such a long signal period will lead to drastic decreasing of the flow, it is not desirable.

5. Conclusion

We studied the effect of traffic signals on vehicle flow in a cellular automata model. In the special case of ASEP (asymmetric simple exclusion process), three distinct regions can be observed in a 3D traffic flow J versus vehicular density ρ and signal period T plot: low and high density oscillations and intermediate density plateau. The introduction of stochastic noise reduces the oscillations and lowers the traffic flow. The influence of the signal period on traffic flow is also greatly reduced. With the increase of speed limit, the traffic flow rises as expected and the high density region expand while the low density region shrinks, resulting in an asymmetry between the low and high density regions, as opposed to the ASEP case. It is interesting to note the scaling behavior of green phase ratio α and lattice size L . To achieve a high traffic flow under fixed density and stochastic noise, a range of signal periods can be used, with lower period corresponding to generally higher flow. For the low and high density regions well separated by the plateau region, two different signal periods can be used to sustain a high flow in their respective region. For two traffic lights on the lattice, the synchronization shift between the two lights

has great influence on the traffic flow, and has to be carefully chosen to maintain a high flow. Of special interest is the appearance of a crater-like structure in the high density region when the shift is small. The reduction of traffic flow is greatest when the two traffic lights are simultaneously switched, i.e., without delay, and is to be avoided.

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